

# Constanza Avalos

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[github.com/cotypolis](https://github.com/cotypolis)

## RESEARCH INTERESTS

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Causal inference models, experimental design, complex survey design, epidemiological research

## RESEARCH STATEMENT

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Robust public health decision-making increasingly relies on the synthesis of diverse data sources, from controlled trials to large-scale observational surveys. My research focuses on developing and applying rigorous causal inference and machine learning frameworks to bridge the evidence gap between randomized controlled trials and real-world populations. Key applications of my work include generalizing the effects of public health interventions from trials to target populations, estimating the real-world impact of policy changes, and developing methods for robust confounding adjustment in high-dimensional observational data using doubly robust machine learning estimators.

## RESEARCH EXPERIENCE

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**University of Manchester** Manchester, United Kingdom  
*PhD Researcher* Oct 2021 — Present

- Implemented and validated doubly robust machine learning estimators (Bayesian additive regression trees and Gradient boosting) to optimize covariate selection for robust confounding adjustment in observational data.
- Designed and executed an online randomized controlled trial (RCT) with a representative sample of 498 UK adults to test food label effects (ATE & CATE).
- Harmonized and curated multiple complex, longitudinal/cross sectional UK nutritional surveys, performing feature engineering on the resulting high-dimensional data to enable robust integration with experimental trial results.

**Alan Turing Institute** London, United Kingdom  
*Enrichment Placement Student* Jan — Jul 2025

- Developed causal inference frameworks to enhance the external validity of clinical trials, transporting causal effects by fusing trial data with observational population data using statistical matching and reweighting techniques (augmented inverse probability weighting, G-computation).
- Deployed advanced ordinal logistic regression models (cumulative link partial proportional odds) to model the evolving association between food label readability and consumption using the UK Food and You Survey (2012-2018).

**University of Manchester** Manchester, United Kingdom  
*Research Assistant* April 2024 — April 2025

- Designed and executed a systematic literature review on the online sale of drugs, performing data extraction and quality assessment on relevant studies.
- Synthesised findings to produce reports detailing key trends, research gaps, and public health implications for policy makers.

## LEADERSHIP & DATA ANALYSIS EXPERIENCE

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**National Institute of Statistics** Santiago, Chile  
*Chief, Department of Agriculture Statistics* April 2018 — January 2019

- Directed the national agricultural statistics program, leading a multidisciplinary team of 15 statisticians, data analysts, and IT staff with a \$1M USD annual budget.
- Oversaw the production and dissemination of 18 institutional publications, including high-frequency (monthly, quarterly, and annual) statistical reports for 12 longitudinal surveys and the 2021 Agricultural Census.
- Managed the successful transition to digital data capture for the 2021 Agricultural Census, which involved implementing a new platform designed for offline data collection in remote rural locations.

*Coordinator, Department of Agriculture Statistics* April 2015 — March 2018

- Oversaw the end-to-end coordination of 6 longitudinal, nationally representative surveys across key agricultural domains—including livestock, crops, and the food industry—all featuring complex multistage sampling designs.
- Led a large-scale field operation, managing and training a team of over 20 supervisors and 100+ surveyors to ensure high-quality data collection across the country.
- Spearheaded the transition to digital data capture, implementing a new platform for offline collection in rural areas that optimized field operations.
  - Reduced the data collection lifecycle from three to two months for 95% of entry points.
- To enhance data integrity, developed and deployed algorithms that flagged inconsistencies at the point of entry using predictive time-series models.
  - This new system resulted in a 10% reduction in measurement error and data missingness.

*Previous experience (2009-2015):* Held data analyst roles focused on program impact evaluation at the **Chilean Ministry of Art and Culture** and consumer data analytics in the private sector.

## EDUCATION

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| <b>University of Manchester</b> , Manchester, United Kingdom<br>PhD in Social Statistics<br>Thesis Title: Causal statistical framework to integrate experimental and observational data to generalize food labeling effects. | Oct 2021 – Dec 2025 (expected submission: Jan 2026) |
| <b>University of Manchester</b> , Manchester, United Kingdom<br>MSc Social Research Methods and Statistics                                                                                                                   | Oct 2020 — Sept 2021<br>Distinction                 |
| <b>Universidad Santiago de Chile</b> , Santiago, Chile<br>BA in Economics                                                                                                                                                    | Mar 2013 — Jul 2016<br>Distinction                  |
| <b>Universidad de Valparaíso</b> , Valparaíso, Chile<br>BA in Sociology                                                                                                                                                      | Mar 2003 — Dec 2008<br>Distinction                  |

## TEACHING & MENTORING

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| <b>University of Manchester</b><br><i>Graduate Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                         | Manchester, United Kingdom | 2023 – 2025 |
| <ul style="list-style-type: none"> <li>Assisted with the delivery of the following postgraduate modules, leading seminars and providing student feedback:             <ul style="list-style-type: none"> <li>SOST70032: Complex Survey Designs and Analysis</li> <li>SOST70172: Quantitative Evaluation of Policies, Interventions and Experiments</li> <li>SOST70520: Methodology and Research Design</li> </ul> </li> </ul> |                            |             |
| <i>Helpdesk Support Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                    |                            |             |
| <ul style="list-style-type: none"> <li>Provided one-on-one technical guidance and methodological troubleshooting to postgraduate students on advanced topics including statistical modelling, complex survey design, and data analysis in R.</li> </ul>                                                                                                                                                                       |                            |             |

## CONFERENCES AND TALKS

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| <b>Jul 2025</b><br><i>PhD Research Showcase, Alan Turing Institute</i>                                       | London, United Kingdom     |
| <b>Jul 2025</b><br><i>SoST RAGS seminar at the Department of Social Statistics, University of Manchester</i> | Manchester, United Kingdom |
| <b>May 2024</b><br><i>Geneva Health Forum, University of Geneva</i>                                          | Geneva, Switzerland        |
| <b>Sept 2023</b><br><i>Royal Statistical Society International Conference</i>                                | Harrogate, United Kingdom  |
| <b>Jun 2022</b><br><i>Postgraduate Summer Research Showcase, University of Manchester</i>                    | Manchester, United Kingdom |

## GRANTS AND AWARDS

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| <b>Grassroots Training Fund</b> , Alan Turing Institute<br><i>Secured funding to co-organize the workshop: "Digital Twins: A collaborative workshop for practical application and knowledge exchange."</i> | £2,000<br>Jul 2025 |
| <b>Enrichment Student Funding</b> , Alan Turing Institute<br><i>Awarded funding to attend the Oxford Machine Learning Summer School (MLx Generative AI).</i>                                               | £1,000<br>Jun 2025 |
| <b>ELLIS Summer School Scholarship</b> , University of Manchester<br><i>Awarded a scholarship to attend the ELLIS Summer School on Machine Learning for Healthcare and Biology.</i>                        | Jun 2024           |
| <b>Student Bursary</b> , Manchester Jean Monnet Centre of Excellence<br><i>Awarded to support a poster presentation at the Geneva Health Forum.</i>                                                        | May 2024           |
| <b>Departmental Bursary</b> , School of Social Sciences, University of Manchester<br><i>Awarded to support a poster presentation at the Royal Statistical Society International Conference.</i>            | Sept 2023          |
| <b>Fieldwork Bursary</b> , School of Social Sciences, University of Manchester<br><i>Awarded to fund primary data collection for an online RCT. Prolific Ltd as a recruiter and data collector.</i>        | £3,000<br>Nov 2022 |
| <b>PhD Fellowship</b> , ANID (Chilean National Research and Development Agency)<br><i>Awarded a fully-funded four-year fellowship for doctoral studies</i>                                                 | Sept 2021          |
| <b>Master's Fellowship</b> , ANID (Chilean National Research and Development Agency)<br><i>Awarded a fully-funded fellowship for master's studies</i>                                                      | Oct 2020           |

## SERVICE

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| <b>University of Manchester</b><br><i>Social Stats Hackathon 2023 organizer</i> | Manchester, United Kingdom<br>Jul 2023 — Sept 2023 |
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University of Manchester  
PGR student representative

Manchester, United Kingdom  
Sept 2023 — Sept 2024

### Entrepreneurial Project (TRAILACADEMY.RUN)

Project Designer & Developer

Manchester / York, UK  
Jul 2024 – Present

- Led the design and business validation for *TRAILACADEMY.RUN*, a proposed AI-driven service using a 'Digital Twin' for trail runner performance optimization.
- Developed a proof-of-concept for modelling Training Stress Score (TSS) by applying practical skills from the 'Neural Networks for Researchers' course (Univ. of York, Nov 2024).
- Implemented predictive models using hands-on training in ML frameworks (e.g., Keras/TensorFlow via Google Colab) and validated the concept with user interviews.

## TECHNICAL SKILLS

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**Methodologies:** Causal Inference (Transportability / Generalizability, G-computation, Augmented Inverse Probability Weighting), Doubly Robust Machine Learning (Bayesian additive regression trees and Gradient boosting), Complex Survey Design & Analysis, Experimental Design (Online RCTs), Longitudinal Data Analysis, Systematic Reviews, Data wrangling

**Software:** R & Python, Latex

## REFERENCES

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### Dr. Yan Wang

Lecturer in Social Statistics, Department of Social Statistics, University of Manchester, United Kingdom

E-mail: yan.wang-5@manchester.ac.uk

Scholar Profiles: Personal Page — Google Scholar — LinkedIn

### Dr. Nick Shryane

Lecturer in Social Statistics, Department of Social Statistics, University of Manchester, United Kingdom

E-mail: N.Shryane@manchester.ac.uk

Scholar Profiles: University of Manchester - Personal Page — Google Scholar

## PUBLICATIONS & REVIEWER EXPERIENCE

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### Journal papers

- **Avalos, C.**, Wang, Y., & Shryane, N. (Manuscript in preparation) On the selection of covariates for transportability: A doubly robust machine learning approach for generalizing a food labelling trial with mixed-type data.
- **Avalos, C.**, Wang, Y., & Shryane, N. (Under review at Journal Appetite). Modelling the evolving association between perceived food label readability and ready-to-eat meal consumption in the UK (2012-2018).
- **Avalos, C.** (In Press). Food label granularity and working memory: effects on food choice in a randomized controlled trial. Journal of Health, Population and Nutrition.

### Technical reports & institutional publications

- National Institute of Statistics (INE), Chile. (2018). *Statistical Yearbook 2018*. INE.  
Role: Authored the chapter on National Agricultural Statistics, providing the definitive annual data and analysis for the sector.
- National Institute of Statistics (INE), Chile. (2018). *Bulletin of Livestock, Crop, and Dairy Food Industry Statistics*. INE.  
Role: Directed the bulletin's production as head of the responsible department, overseeing the analysis and dissemination of high-frequency agricultural data.
- Food and Agriculture Organization of the United Nations (FAO). (2019). *World Programme of the Census of Agriculture 2020: Operational Guidelines*. FAO.  
Role: Co-authored the Chilean case study for the global guidelines, detailing innovations in national survey design, sampling, and digital data collection methodologies.

### Peer review

- Reviewed manuscripts for the British Food Journal.